Chang Xu

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3 George Julius Ave., Zetland, Sydney, NSW 2017, Australia

Education

- **The University of New South Wales** *Sep 2023 - Present*
  PhD - Climate Science Sydney, Australia
  - Project. The Role of Moisture Fluxes in the Propagation of the Convectively Coupled Waves
  - Supervisors. Prof. Steven Sherwood, Dr. Martin Jucker, Dr. Clemente Lopez Bravo
  - Professional Memberships. UNSW, Climate Change Research Centre (CCRC) (Dec. 2024 - Present); 21st Century Weather (Dec. 2024 - Present); The ARC Centre of Excellence for Climate Extremes (CLEX) (Sep. 2023 - Feb. 2025)

- **Sun Yat-sen University** *Sep. 2019 - Jun. 2022*
  Master of Engineering - Hydraulic Engineering Zhuhai, P.R.China
  - Project. The Spatiotemporal Distribution and Evolution Characteristics of Heavy Rainfalls and Integrated Water Vapor Transport in Mainland China
  - Supervisor. Prof. Zhihe Chen
  - GPA: 3.81/4.00

- **University of Jinan** *Sep. 2015 - Jun. 2019*
  Bachelor of Engineering - Hydrology and Water Resources Engineering Jinan, P.R.China
  - GPA: 3.41/4.00

Publications

J=Journal, S=In Submission, T=Thesis, P=Patent, R=Report

[J.1] Chang Xu, Steven Sherwood, Martin Jucker, Clemente Lopez Bravo, Richard Jones (2026). **The Evaluation of Convectively Coupled Waves in Limited-Length Datasets**. *Climate Dynamics*, 64(145). https://doi.org/10.1007/s00382-026-08106-2

[J.2] Yuanrui Chen, Chang Xu, Steven Sherwood, and Jonathon S. Wrighta (2026). **Interactions Between the SPCZ and Convectively Coupled Equatorial Waves**. *Journal of Climate*, Under Review.

[J.3] Chang Xu, Zhihe Chen, and Zhihua Zhu (2022). **Spatiotemporal Patterns and Evolution of Heavy Rainfall Trajectories in China**. *International Journal of Climatology*, 42(16), 8632–8651. https://doi.org/10.1002/joc.7754

[S.1] Chang Xu, Steven Sherwood, Martin Jucker, Clemente Lopez Bravo (2026). **Robustness of Madden-Julian Oscillation Diagnostics Across Global Precipitation Datasets**. *Geophysical Research Letters*, under review.

[T.1] Chang Xu (2022). **Research on the spatiotemporal distribution and evolution characteristics of heavy rainfalls and integrated water vapor transport in mainland China (in Chinese)**. School of Civil Engineering, Sun Yat-sen University, 102pp.

[T.2] Chang Xu (2019). **Analysis of surface runoff changes and causes in Xiaoqing River Basin, Jinan City (in Chinese)**. School of Water Conservancy and Environment, the University of Jinan, 29pp.

[R.1] Chang Xu (2026). **The gender blind spot in climate science: why our models overlook gender inequity**. *Climate Perspectives (pp. 149)*, 64(145). The Climate Centre

[P.1] Chang Xu, Zhihe Chen, and Zhihua Zhu (2021). **Integrated water vapor transport channel analysis calculation and visualization system (v1.0)**. China patent, 2021SR1596525.

Presentations and Conferences

P=Presentation, C=Conference

[P.1] Chang Xu (2026). The Phase-dependent Moisture-Convection Coupling of the MJO. In *Circulation Workshop 2026*, Melbourne, Austrialia.

[P.2] Chang Xu (2026). The Evaluation of Convectively Coupled Waves in Limited-Length Datasets. In *AMOS 2026*, Hobart, Austrialia.

[P.3] Chang Xu (2026). The Role of Moisture Fluxes During the Propagation of the Convectively Coupled Waves (CCWs). In *MPI-M Joint Seminar*, Hamburg, Germany.

[P.4] Chang Xu, Zhihe Chen, and Zhihua Zhu (2019). **Study on the intra-annual spatial and temporal distribution of heavy rainfall in China and its evolutionary characteristics**. In *18th China Water Forum*, Nanjing, Jiangsu, P. R. China.

Additional Research Experience

- **Visiting researcher: Moisture Budget in ICON-CPM** *Jan. 2026 - Feb. 2026*
- **Co-author: Convectively-Coupled Systems in the tropics** *May 2025 - Present*
- **Research assistant: Aggregation of fine-grained sediment and microplastics in estuaries in Dongjiang China** *Feb. 2022 - Mar. 2022*
- **Research assistant: Hydro-sand dynamics and its ecological effects in Jinsha River** *Jan. 2021 - Jun. 2021*
- **Research assistant: Water Resources Utilization and Development in Cambodia** *Nov. 2020 - Jan. 2021*
- **Project leader: Research on the role of plants in water conservation** *Sep. 2019 - Jun. 2022*

Honors and Awards

- **Development and Research Training Grant (DRTG)** *Apr. 2025 - Present*
  The University of New South Wales
- **University International Postgraduate Award (UIPA)** *Sep. 2023 - Present*
  The University of New South Wales
- **Science - CCRC funded Top-ups** *Sep. 2023 - Present*
  UNSW CCRC
- **Second-class scholarship for graduate students** *2021 - Feb. 2022*
  Sun Yat-sen University
- **Outstanding Thesis of Jinan University** *Jun. 2019*
  The University of Jinan
- **First-class scholarship of the University of Jinan** *Jun. 2019*
  The University of Jinan
- **Third-class scholarship of the University of Jinan (three times)** *2015 - 2018*
  The University of Jinan
- **The Challenge Cup National College Students' Extracurricular Academic** *Feb. 2017 - Apr. 2017*
  The University of Jinan
- **Science and Technology Works Contest. Third prize** *Feb. 2017 - Apr. 2017*
  The University of Jinan
- **Outstanding Student of the Year** *Nov. 2016*
  The University of Jinan
- **Second-class scholarship of the University of Jinan** *Nov. 2016*
  The University of Jinan

Academic Engagements

- **the ARC Centre of Excellence for Weather of the 21st Century's annual Graduate Colloquium** *May 2025 & Jun 2026*
  21st Century Weather
- **Warmer World and Variability Meeting** *May 2025*
  21st Century Weather
- **The CLIVAR Climate Dynamics Panel (CDP) Annual workshop** *Feb. 2025*
  The ARC Centres of Excellence for Climate Extremes (CLEX) and 21st Century Weather
- **CLEX Annual workshop 2024** *Dec. 2024*
  The ARC Centres of Excellence for Climate Extremes (CLEX) and 21st Century Weather
- **2024 Annual Research and Development workshop** *Sep. 2024*
  The Bureau of Meteorology
- **Annual Winter School_Weather and Climate Interactions** *Jun. 2024*
  The ARC Centres of Excellence for Climate Extremes (CLEX) and 21st Century Weather
- **CLEX Annual workshop 2023** *Dec. 2023*
  The ARC Centres of Excellence for Climate Extremes (CLEX)

Skills

- **Programming Languages:** Python, Matlab, Linux
- **Data Science & Machine Learning:** Python (NumPy, Pandas, xarray, scikit-learn), MATLAB, Jupyter, Git
- **Specialized Area:** Atmospheric Dynamics, Moisture Budget Analysis, MJO & Equatorial Waves, Climate Modeling
- **Mathematical & Statistical Tools:** Fourier Analysis, Wavelet Transform, AR(1) Modeling, Bootstrapping, Regression, Hypothesis Testing
- **Other Tools & Technologies:** Linux (Bash), GitHub, NetCDF, HDF5, LaTeX, NCL, Cartopy, Matplotlib
- **Research Skills:** High-Resolution Model Evaluation, Time Series Analysis, Signal Detection, Data Visualization, Scientific Writing, Proposal Development

Additional Information

Languages: Chinese (Native Language), English (IELTS: 7.0), German (Reading only)
Interests: Moisture-Convection Coupling, Climate Dynamics, Equatorial Wave Theory, Scientific Computing, Data Visualization, Science Communication